

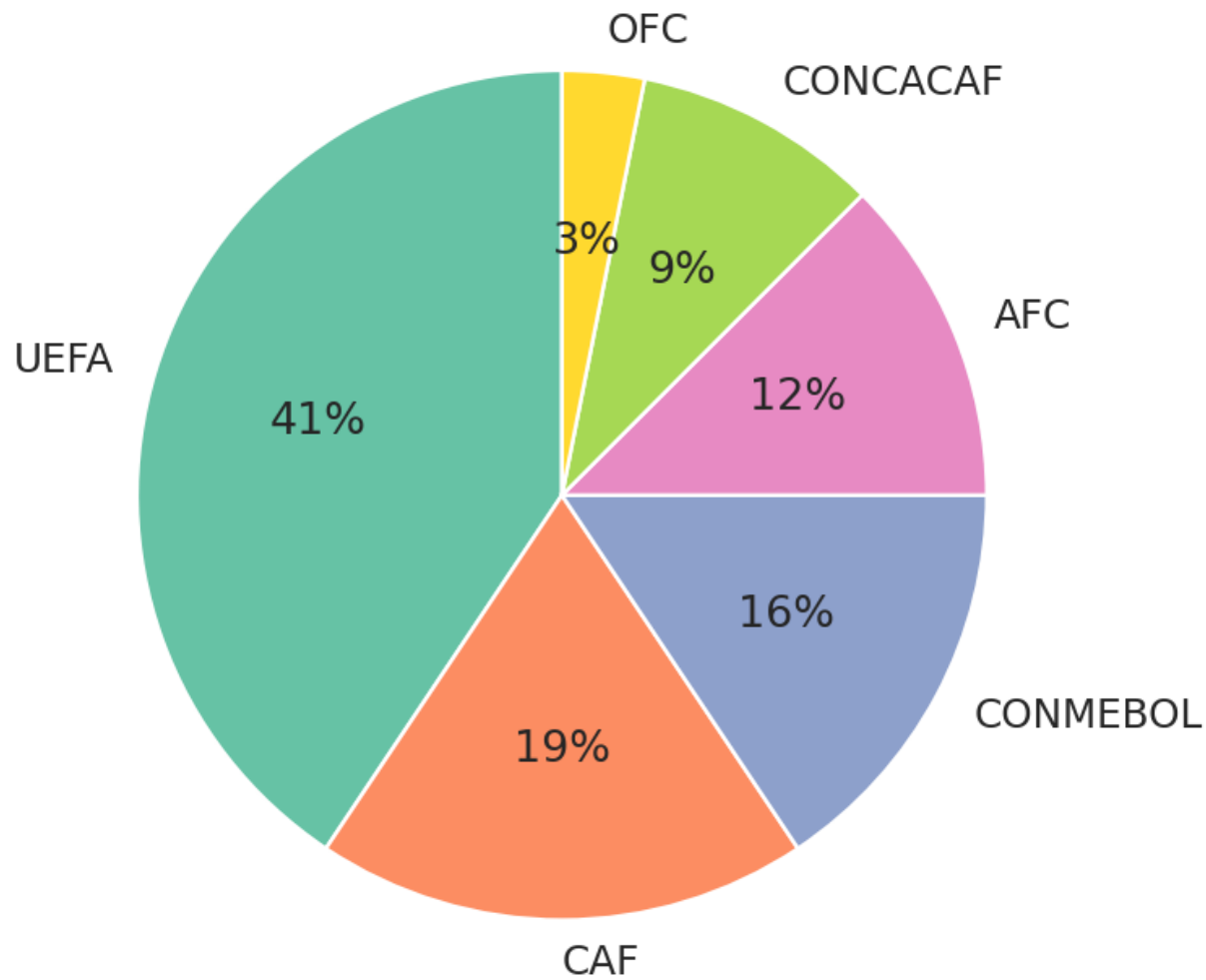
World Cup Standings Explorer

A Visual & Machine-Learning Report

This report accompanies the **team_standings.csv** dataset (32 teams, final Standing + Team name). It presents 14 charts covering classic visualizations -- pie, bar, horizontal bar, line, scatter, heatmap / correlation, geolocation, boxplot -- and machine-learning techniques -- linear regression, classification (Perceptron), prediction, DBSCAN clustering, and hierarchical clustering -- each with a short description placed beside it.

Note: the source file only contains **Standing** and **Team**. Confederation and capital coordinates shown here are real. All performance metrics ending in *_sim* (points, goals, possession, pass accuracy, shots on target) are **simulated** -- generated with a fixed random seed and correlated with final standing -- purely to demonstrate these charts and models. They are not the actual 2010 World Cup match statistics.

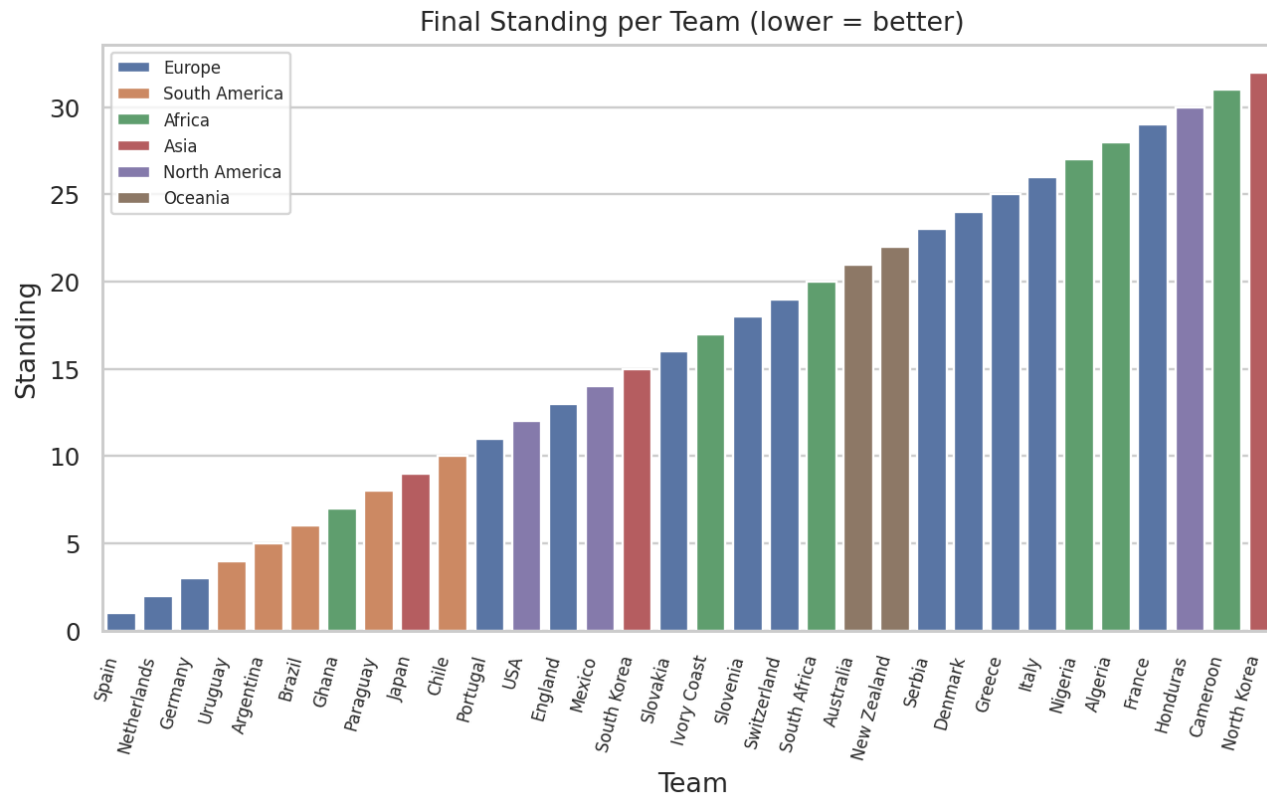
Teams per Confederation



PIE CHART

Teams per Confederation

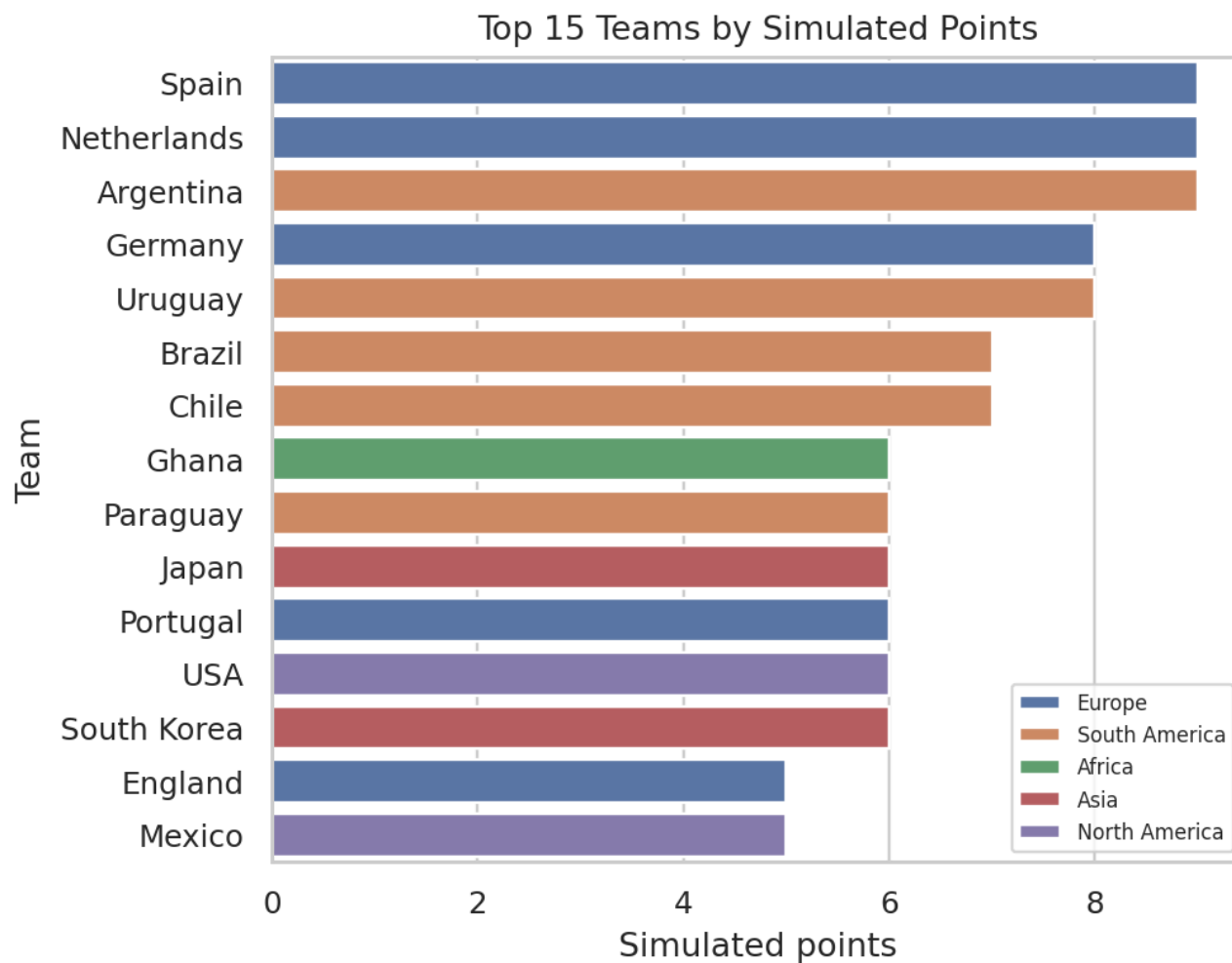
Shows the share of the 32 qualified teams coming from each football confederation. UEFA (Europe) sends by far the largest contingent, which is why it dominates this pie chart.



BAR CHART

Final Standing per Team

Every team ordered by where it finished, from 1st (Spain) to 32nd (North Korea). Bars are colored by continent so you can spot which regions clustered near the top versus the bottom of the standings.

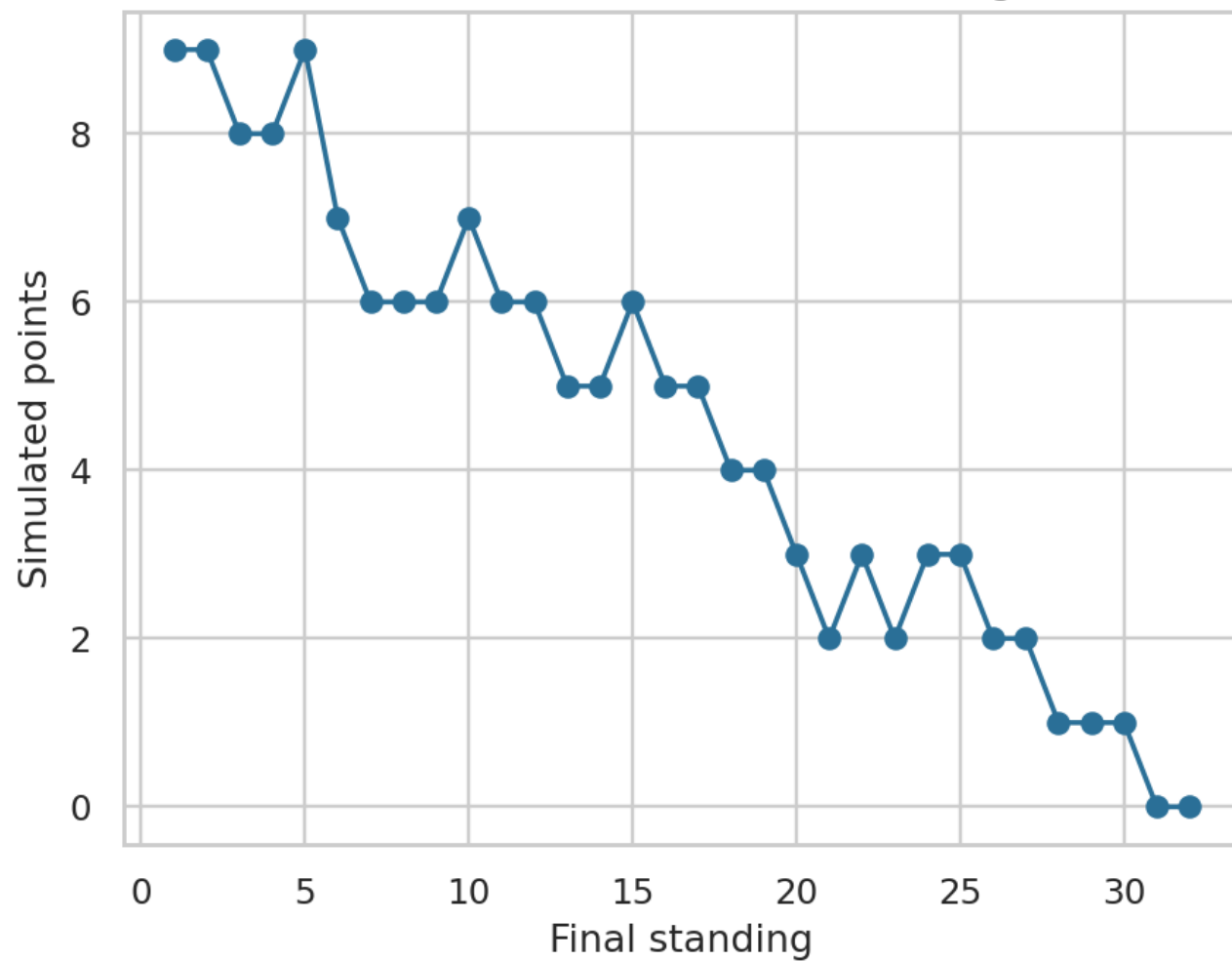


HORIZONTAL BAR CHART

Top 15 Teams by Simulated Points

The 15 highest scorers on the simulated points metric, drawn as a horizontal bar so long team names stay readable. The top of this points ranking broadly agrees with the real final standing.

Simulated Points vs. Final Standing

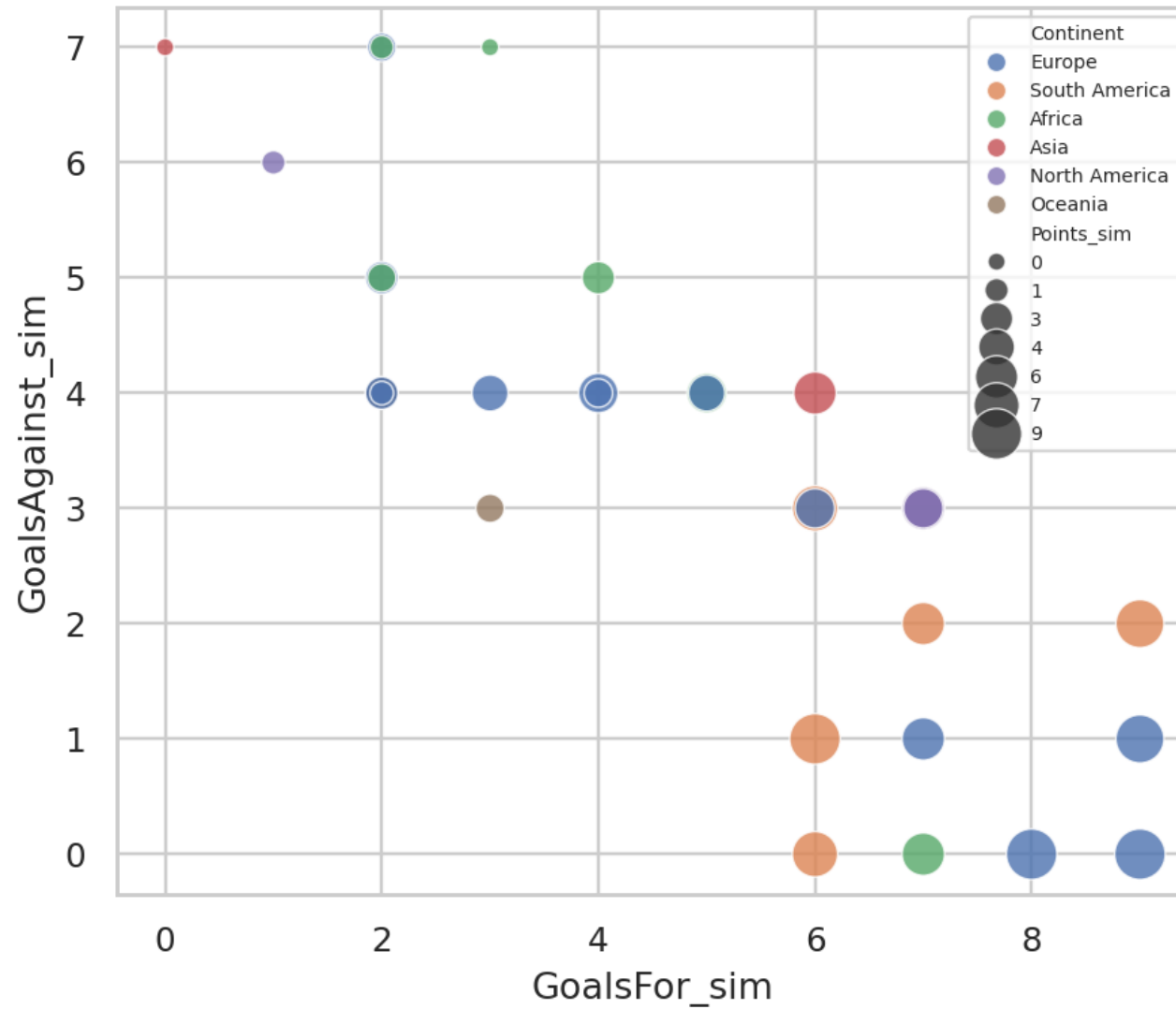


LINE CHART

Simulated Points vs. Final Standing

Traces simulated points as standing goes from 1st to 32nd. The downward trend (with some noise) is expected by construction: points were simulated to correlate with final standing.

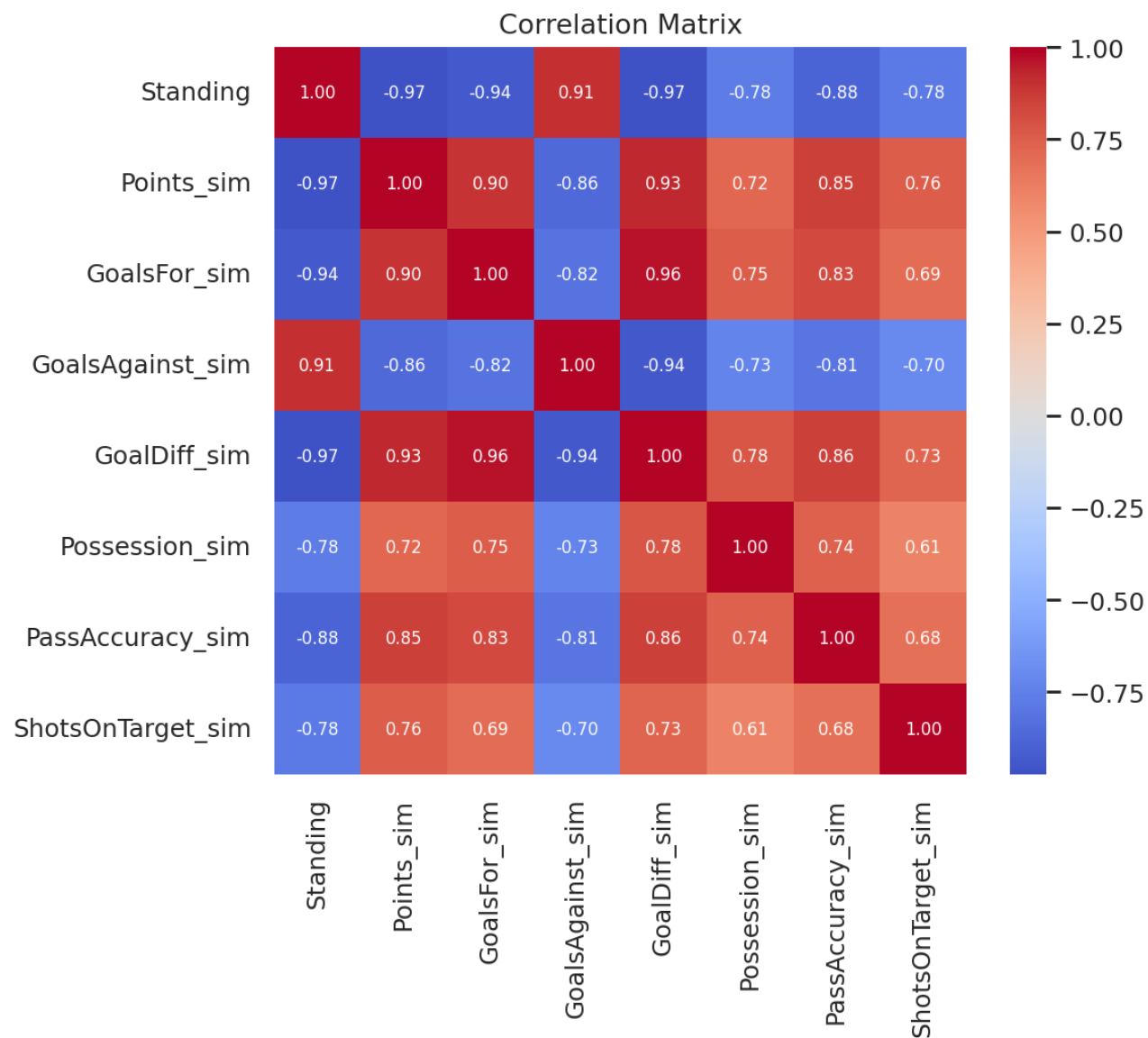
Goals For vs. Goals Against (bubble = points)



SCATTER PLOT

Goals For vs. Goals Against

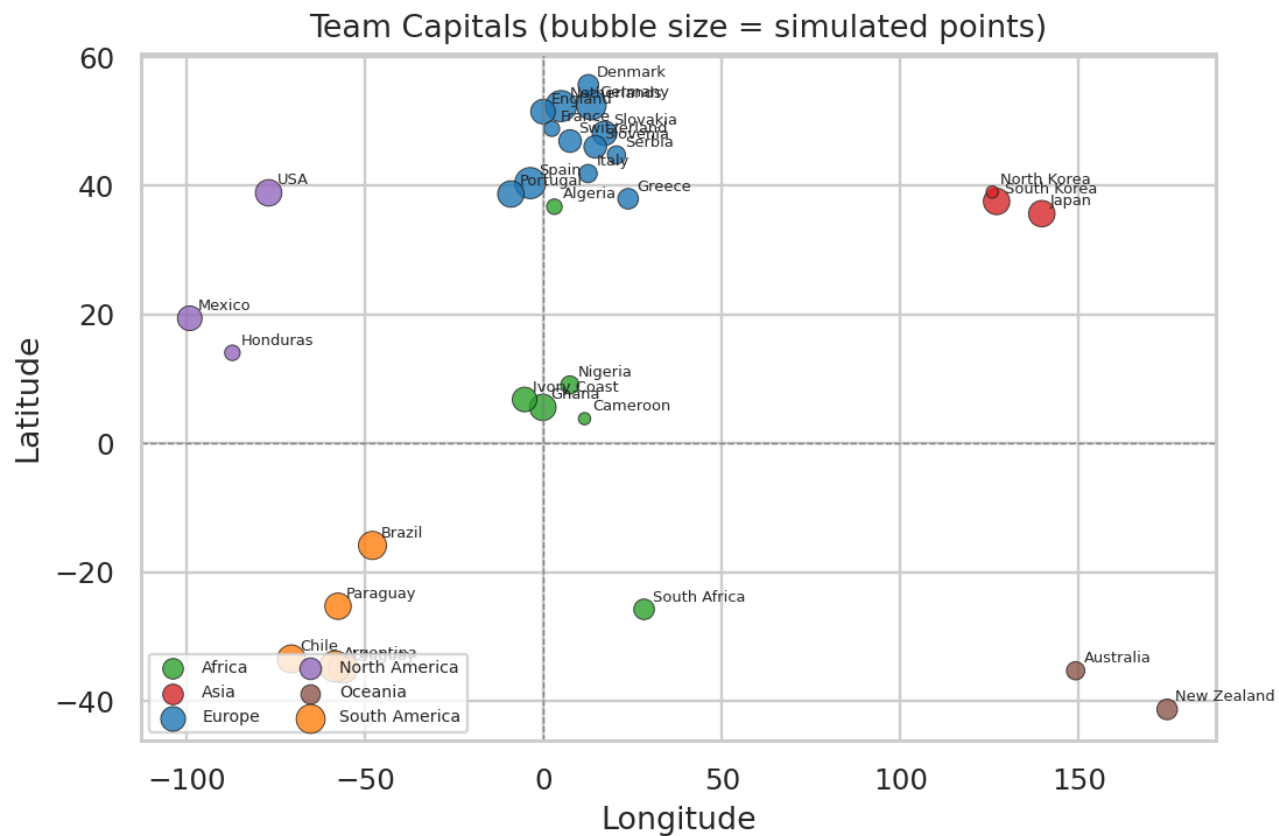
Each bubble is a team. Teams in the bottom-right (many goals for, few against) scored the most simulated points -- shown by bubble size -- and tended to finish higher in the standings.



HEATMAP / CORRELATION

Correlation Matrix of Numeric Features

Pearson correlation between every numeric column. Values near +1/-1 (deep red/blue) mean two columns move strongly together or in opposite directions; values near 0 mean little linear relationship.

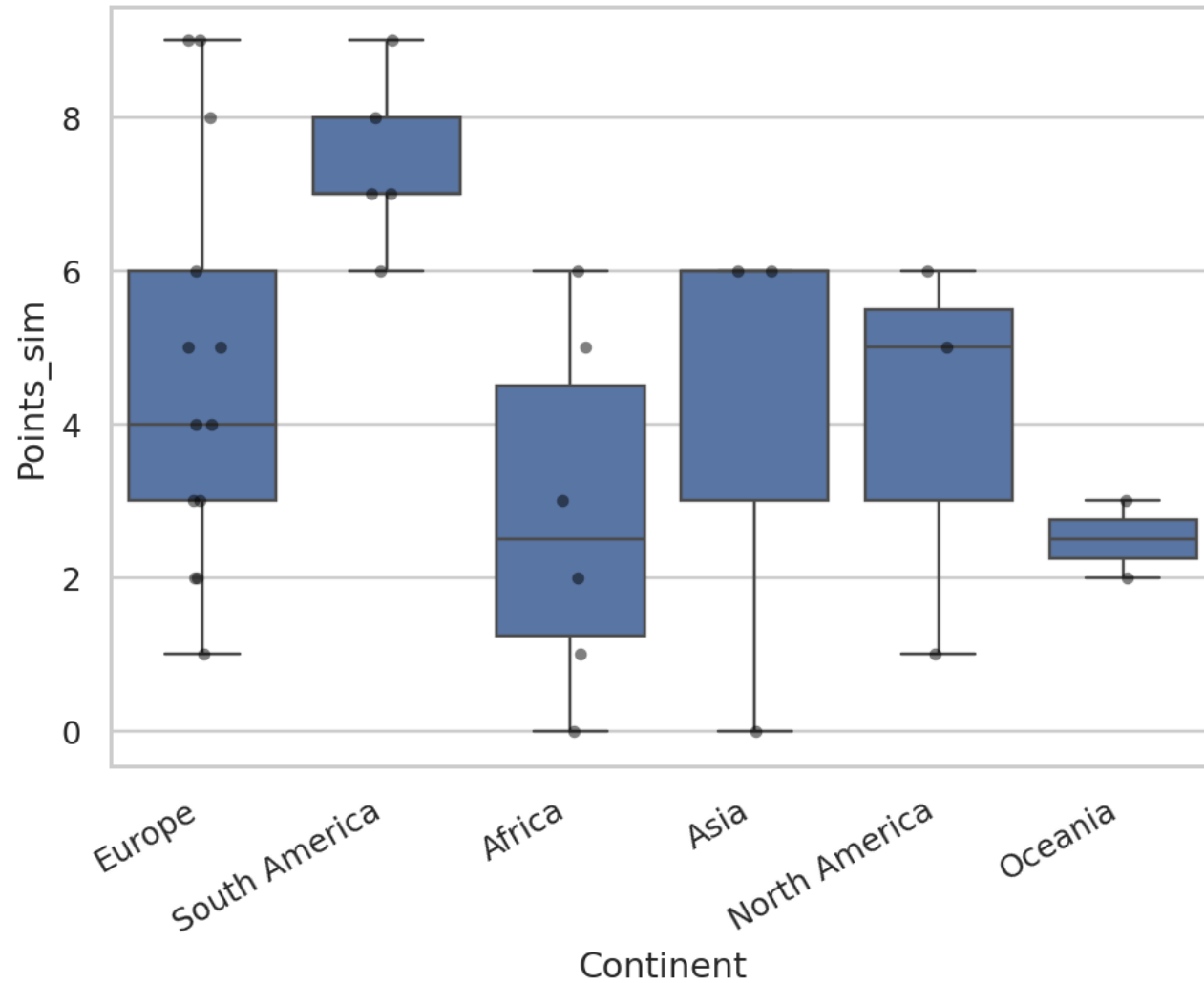


GEOLOCATION

Team Capitals on a World Map

Each team is placed at its real capital-city coordinates. Bubble size encodes simulated points, and color encodes continent, making it easy to see which regions of the world performed best.

Simulated Points Distribution by Continent

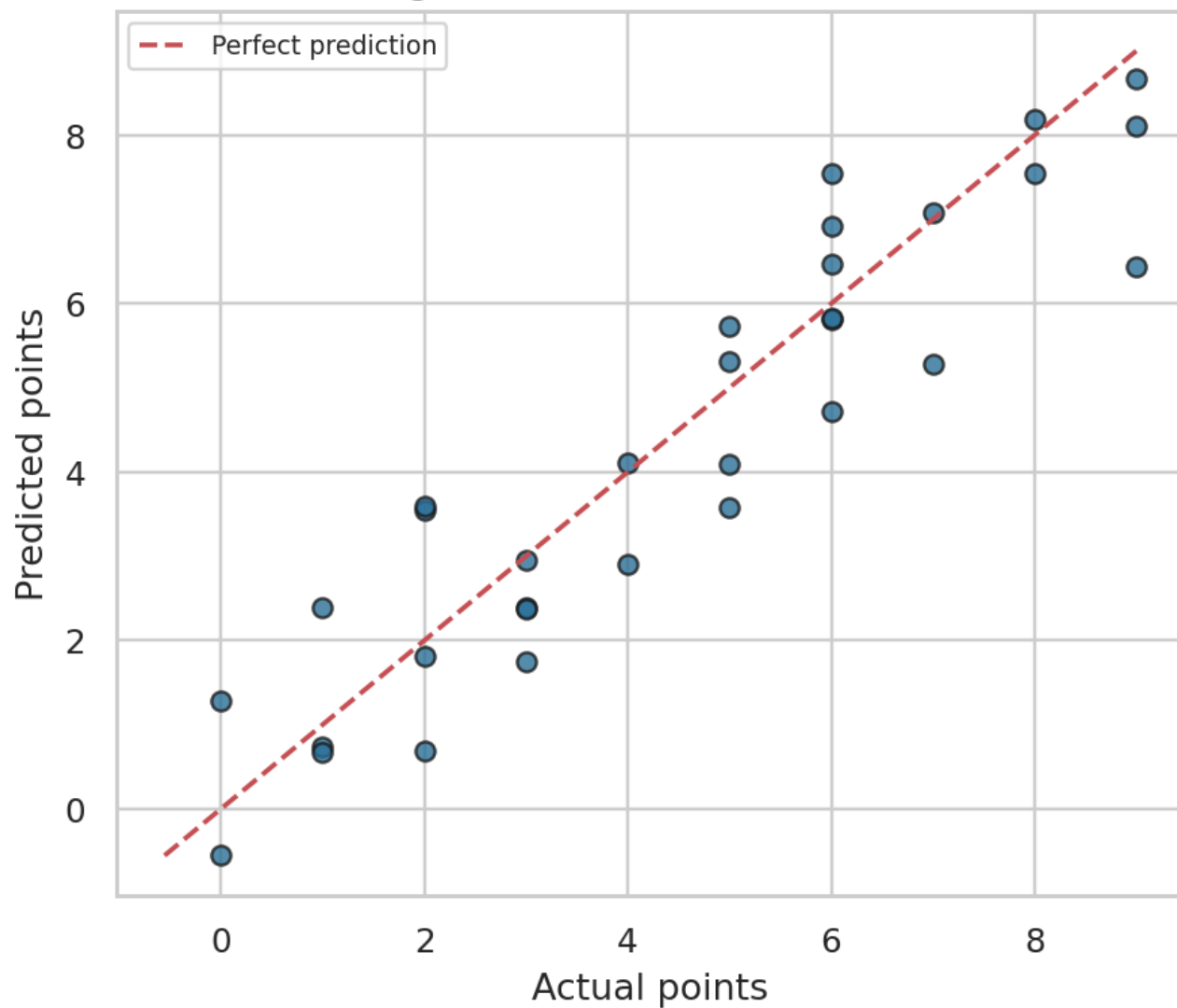


BOXPLOT

Simulated Points by Continent

The box spans the middle 50% of values, the line inside is the median, and individual dots are the actual teams -- useful for spotting outliers and comparing spread across continents.

Linear Regression: Actual vs. Predicted Points

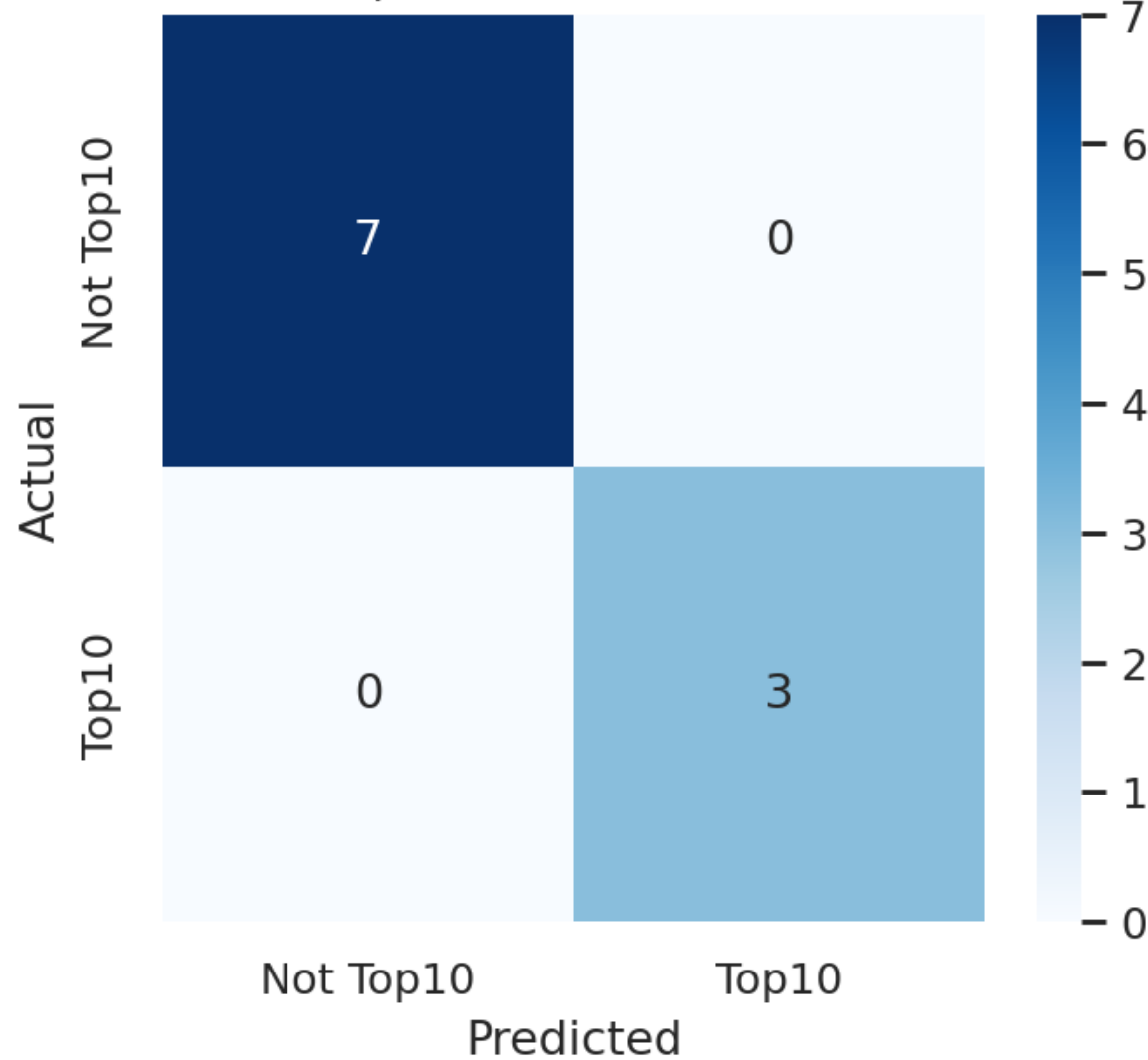


LINEAR REGRESSION

Actual vs. Predicted Points

A scikit-learn LinearRegression model predicts simulated points from goal difference and possession. Points sitting exactly on the red dashed line would be perfect predictions; the tighter the cloud hugs the line, the better the fit.

Perceptron Confusion Matrix

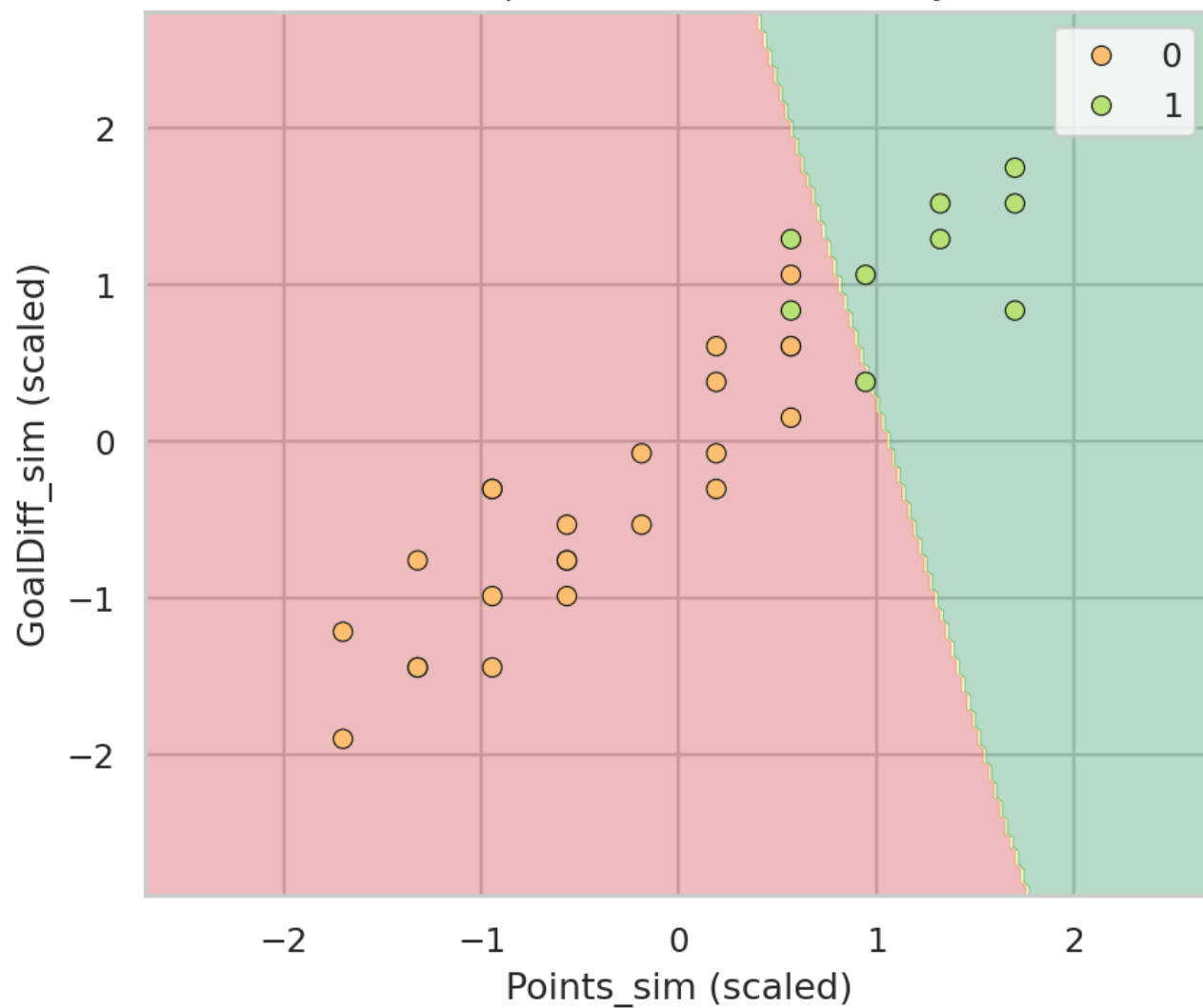


CLASSIFICATION (PERCEPTRON)

Confusion Matrix

A Perceptron classifier predicts whether a team finished in the Top 10. Rows are actual classes, columns are predicted classes -- diagonal cells are correct predictions, off-diagonal cells are mistakes.

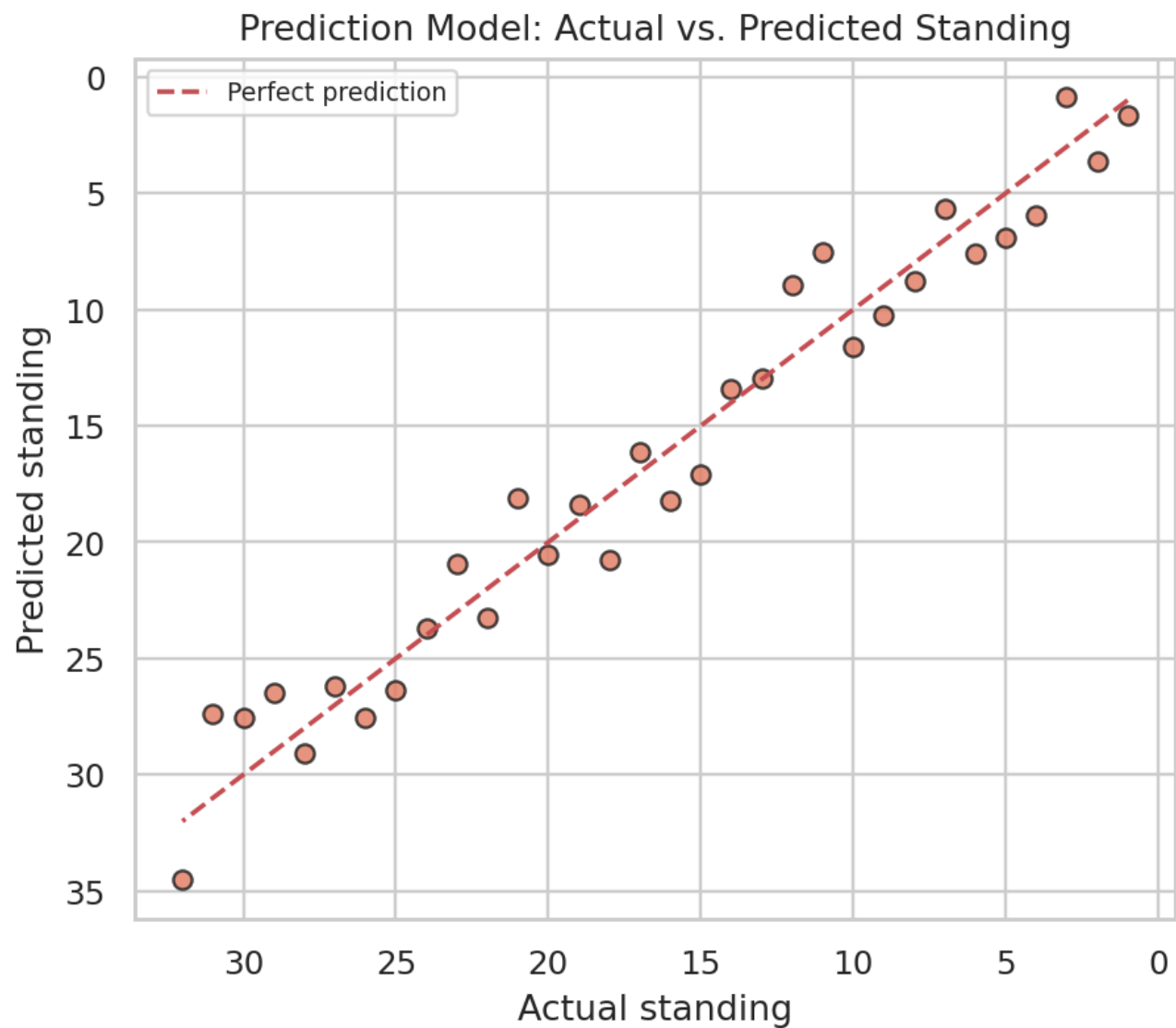
Perceptron Decision Boundary



PERCEPTRON DECISION BOUNDARY

Top 10 vs. Not Top 10

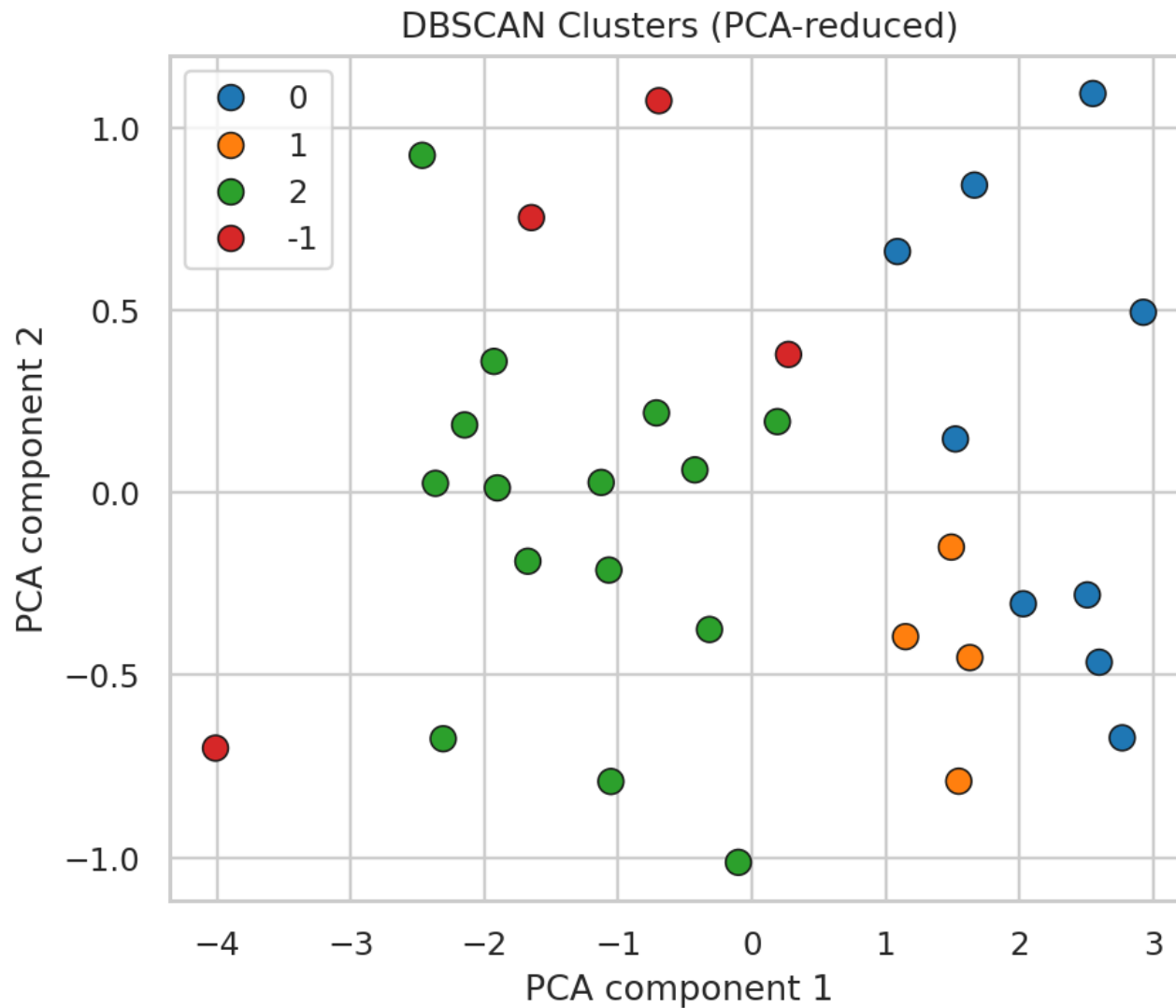
Shows the boundary the Perceptron draws between the two classes using two scaled features (simulated points and goal difference). Dots are real teams colored by their true class.



PREDICTION

Predicted vs. Actual Standing

A regression model trained on simulated match statistics predicts each team's final standing. Axes are inverted so 'better' (lower standing number) is toward the top-right, matching intuition.

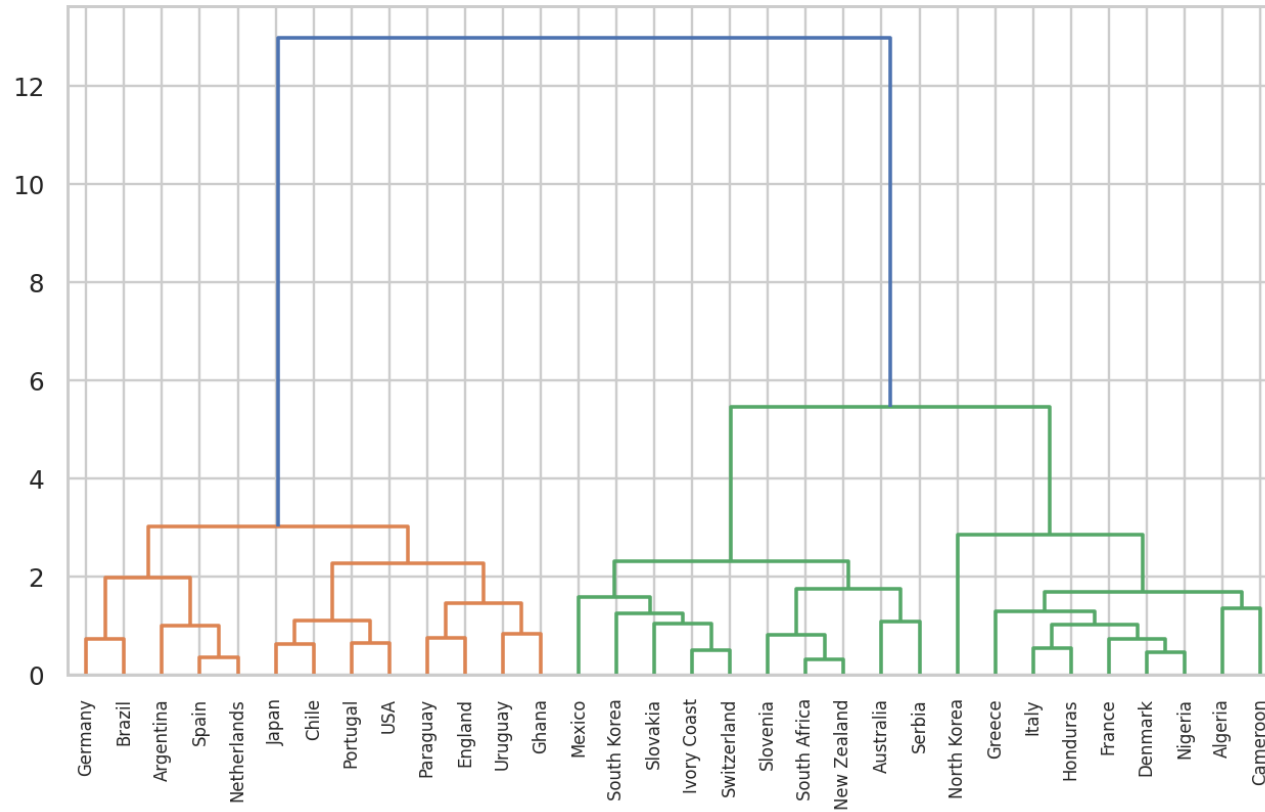


DBSCAN CLUSTERING

Density-Based Clusters (PCA-reduced)

Teams projected onto two principal components and colored by DBSCAN cluster. Cluster '-1' marks teams DBSCAN considered noise -- they did not fit densely into any group.

Hierarchical Clustering Dendrogram (Ward linkage)



HIERARCHICAL CLUSTERING

Ward-Linkage Dendrogram

A tree showing how teams merge into groups at increasing distance thresholds. Teams that join together low down (short vertical distance) are the most similar on the clustering features.